

CSE 0613-2203 | Design and Analysis of Algorithms

Topic: Introduction of Algorithms

1. Algorithm

An **algorithm** is a step-by-step procedure that defines a set of instructions to be executed in a certain order to get the desired output.

| Key Points

- Step-by-step procedure
- Produces desired output
- Language independent
- Can be implemented in different programming languages

2. Categories of Algorithms

- **Search:** Finding an item in a data structure.
- **Sort:** Arranging items in a particular order.
- **Insert:** Adding an item to a data structure.
- **Update:** Modifying an existing item.
- **Delete:** Removing an item.

3. Origin of the Word Algorithm

The word **Algorithm** comes from the name of the Persian mathematician:

Abu Ja'afar Mohammed Ibn Mousa Al Khwarizmi (825 A.D.)

He wrote several books on Mathematics and Algebra.

4. Characteristics of an Algorithm

| 1. Unambiguous

Every step must be **clear and have only one meaning**.

| 2. Input

An algorithm can have **zero or more well-defined inputs**.

| 3. Output

An algorithm must have **one or more well-defined outputs**.

| 4. Finiteness

An algorithm must **terminate after a finite number of steps**.

| 5. Feasibility

The algorithm must be possible with the available resources.

6. Independent

An algorithm should be independent of programming code/language.

Remember

U → **I** → **O** → **F** → **F** → **I**

5. Ways to Express an Algorithm

Natural Language

Written using plain English.

- Easy to describe
- May become unclear for complex problems

Flowchart

A graphical representation of algorithm steps.

- Easy to visualize

Pseudocode

Text-based, code-like instructions without specific programming-language syntax.

- Easy to understand
- Considered a good way to express an algorithm

6. Steps to Design an Algorithm

1. **Problem Definition** – Clearly define the problem.
2. **Constraints** – Identify limitations or rules.
3. **Inputs** – Determine the given data.
4. **Outputs** – Specify the expected result.
5. **Solution Feasibility** – Ensure the solution works within the constraints.

Shortcut

Problem → **Constraints** → **Input** → **Output** → **Feasibility**

7. Algorithm Analysis

A Priori Analysis

A theoretical analysis performed before implementation.

- Mathematical analysis
- Assumes other factors are constant
- Uses **Ω** , **O** , and **θ**

A Posteriori Analysis

An empirical analysis performed after implementation.

- Algorithm is implemented and executed

- Actual running time and space are measured
- Example: **0.18 seconds, 8 MB**

Easy Difference

A Priori	A Posteriori
Before implementation	After implementation
Theoretical	Empirical
Mathematical counting	Run and measure
Ω , O , θ	Actual time and space

8. Algorithm Complexity

The two main factors that determine algorithm efficiency are:

Time Factor

Measures the time/number of key operations required by an algorithm.

Example: Number of comparisons in a sorting algorithm.

Space Factor

Measures the maximum memory space required by an algorithm.

Complexity Function

If n is the size of input data, the complexity $f(n)$ represents the running time and/or storage space required by the algorithm in terms of n .

Quick Revision

Most Important Topics

1. Definition of Algorithm
2. Characteristics of Algorithm
3. Categories of Algorithms
4. Origin of Algorithm
5. Natural Language, Flowchart & Pseudocode
6. Steps to Design an Algorithm
7. A Priori vs A Posteriori Analysis
8. Time Factor & Space Factor
9. Algorithm Complexity